Exponential online mistakes at every width for isotropic threshold networks with a residual

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Abstract

We construct a Bayesian neural-network learner that has polynomial IID sample complexity but makes exponentially many mistakes on a fixed input order, at every hidden width in its strong-prior limit. The activations are the fixed discontinuous functions $\mathrm{sign}(t) + \tanh(t)$ and $\mathrm{sign}(t) + t$: a threshold neuron with a bounded or linear residual. The bounded version also works with an arbitrarily small fixed positive coefficient on its tanh residual. The network has one hidden layer, the full Boolean input cube, a constant input coordinate, and the standard independent Gaussian weight prior with equal variances across all first-layer coordinates. The prior is conditioned on global output margin greater than $1/2$ and exact labels. For a target of Boolean circuit size $\Theta(k)$, a calibration prefix of $O(k)$ examples is followed by $2^k - k - 1$ designated mistakes. The target itself has a width-one teacher with every weight bounded by one. An exact all-competitor inequality reduces the minimum Gaussian energy, at every width, to a strictly increasing function of the affine margin norm. A separate convex-hull argument gives an energy gap to all nonthreshold functions, yielding polynomial IID sample complexity. We also prove direct width-dependent variance statements: every fixed exponent $\gamma > 1$ works at all sufficiently large widths, and a sufficiently small mean-field prefactor works uniformly at every width. The proof uses the activation's jump at zero; no conclusion for ordinary tanh, ReLU, or hardtanh is claimed.

1 The learner and the separation

We study two fixed activations, with $\mathrm{sign}(0) = 0$:

$$\phi_{\mathrm{lin}}(t) = \mathrm{sign}(t) + t, \qquad \phi_{\mathrm{bnd}}(t) = \mathrm{sign}(t) + \tanh(t).$$

Both are odd and have a jump at zero. The second is bounded in absolute value by two. Each is a threshold activation with a residual, rather than a continuous approximation to a threshold. In the model below, ϕ may be either one. For $x \in X = \{-1, 1\}^d$, write $\bar{x} = (1, x)$. A width-N learner has output

$$F_{u,v}(x) = \sum_{j=1}^{N} v_j \phi(u_j \cdot \bar{x}), \qquad u_{ji} \sim N(0, N\sigma^2/d), \quad v_j \sim N(0, \sigma^2/N), \tag{1}$$

with all weights independent. The bias is the coefficient of the constant input and has the same variance as every other first-layer coordinate. There is no coordinate-dependent covariance or extra readout regularization factor.

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For a Boolean function g let

$$U_g = \{(u, v) : g(x)F_{u,v}(x) > 1/2 \text{ for every } x \in X\}.$$

The posterior first conditions on the union of these full-cube strict-margin cells and then retains exactly the cells consistent with the observed labels. A prediction's error probability is the posterior mass of functions assigning the wrong label. This is the error probability of a fresh posterior sample. Posterior-label majority is wrong when that mass exceeds $1/2$; the stronger convention requiring correct-label probability greater than $2/3$ is also considered.

The strong-prior limit at a fixed width means $\sigma \downarrow 0$. For direct width-dependent statements we use

$$\sigma_N^2 = \tau^2 N^{-\gamma}, \qquad \mathrm{Var}(u_{ji}) = \tau^2 N^{1-\gamma}/d, \quad \mathrm{Var}(v_j) = \tau^2 N^{-\gamma-1}. \tag{2}$$

Here $\tau > 0$ and $\gamma \geq 1$ are fixed.

Theorem 1. *For every $k \geq 3$, put $d = 2k + 1$ and write inputs as (x, y, z), where $x, y \in \{-1, 1\}^k$ and $z \in \{-1, 1\}$. Define*

$$X_x = \sum_{i=0}^{k-1} 2^i \frac{x_i + 1}{2}, \qquad Y_y = \sum_{i=0}^{k-1} 2^i \frac{y_i + 1}{2}, \qquad f_k(x, y, z) = +1 \iff z = -1 \text{ and } X_x \geq Y_y.$$

There is one fixed permutation of the input cube, depending only on k, with the following properties for either activation.

1. *The target has a Boolean circuit of size $\Theta(k)$ and a width-one teacher for the activation ϕ, with every weight at most one in absolute value and signed output at least one everywhere.*

2. *After at most $4k + 3$ calibration examples, there are $T_k = 2^k - k - 1 \geq 2^{k-1}$ designated rounds. At every fixed width $N \geq 1$, the posterior wrong-label probability at each of these rounds tends to one as $\sigma \downarrow 0$.*

3. *For every fixed $\gamma > 1$ and $\tau > 0$ in (2), the same wrong-label probabilities tend to one as $N \to \infty$.*

4. *For every $\eta \in (0, 1)$ there is $\tau_0(k, \eta) > 0$ such that, for every $0 < \tau \leq \tau_0$, every $\gamma \geq 1$, and every $N \geq 1$, posterior sampling makes all T_k designated mistakes with probability at least $1 - \eta$. The bound also makes each designated posterior majority wrong, and violates the $2/3$ correctness criterion, when η is chosen sufficiently small.*

5. *For the same prior and target, the realizable IID sample complexity is $O((d^2 + \log(1/\beta))/\varepsilon)$ for correct-label posterior probability greater than $2/3$ outside a set of input mass at most ε, with sample confidence at least $1 - \beta$. This holds at each fixed width for sufficiently small σ, and under the direct width limits specified above.*

No choice among tied predictors is used. Every network that would make the correct next prediction is included in the energy comparison.

The discontinuity distinguishes this example from ordinary tanh or ReLU. The theorem keeps the Gaussian prior and its usual layer normalizations unchanged, but changes the activation. The small-prefactor assertion is not a claim about the unit-prefactor case $\gamma = 1, \tau = 1$.

2 An exact minimum over every hidden width

We first prove the simpler statement for $\phi = \phi_{\mathrm{lin}}$. Set $b_j = Nv_j$. The Gaussian rate energy and output become

$$E_N = \frac{d}{N}\sum_j \|u_j\|^2 + \frac{1}{N}\sum_j b_j^2, \qquad F(x) = \frac{1}{N}\sum_j b_j \phi(u_j \cdot \bar{x}). \tag{3}$$

Consider any finite linearly separable labeled set, and put $z_i = y_i \bar{x}_i$. Let a_* minimize $\|a\|^2$ under $z_i \cdot a \geq 1$, and write $Q = \|a_*\|^2 > 0$. The minimizer is unique. The normal-cone condition for this Euclidean projection gives active constraints and nonnegative multipliers with

$$a_* = \sum_i \mu_i z_i, \qquad \mu_i > 0 \Longrightarrow z_i \cdot a_* = 1, \qquad \sum_i \mu_i = Q.$$

The last identity follows by taking the inner product with a_*. Put $\rho_i = \mu_i/Q$.

Theorem 2 (Exact all-width affine reduction)**.** *At every $N \geq 1$, the minimum of (3) over all networks satisfying $y_i F(x_i) \geq 1/2$ is*

$$\Psi_d(Q) = \min_{r \geq 0} \left\{ dQr^2 + \frac{1}{4(1+r)^2} \right\}. \tag{4}$$

Its unique minimizing radius is strictly positive and obeys

$$4dQr(1+r)^3 = 1.$$

The minimum is attained by cloning $u_j = ra_*$ and $b_j = 1/[2(1+r)]$. The function $\Psi_d(Q)$ is strictly increasing in $Q > 0$ and is always strictly less than $1/4$.

Proof. Define the active-data average

$$S(u) = \sum_i \rho_i \phi(z_i \cdot u) = \sum_i \rho_i \operatorname{sign}(z_i \cdot u) + a_* \cdot u/Q.$$

Since the sign average has absolute value at most one,

$$|S(u)| \leq 1 + \|u\|/\sqrt{Q}.$$

Let $\mathbb{E}_j$ mean uniform averaging over the N neurons and set $r^2 = \mathbb{E}_j\|u_j\|^2/Q$. Oddness of ϕ and the training margins give

$$\begin{aligned} \frac{1}{2} &\leq \mathbb{E}_j[b_j S(u_j)] \\ &\leq \sqrt{\mathbb{E}_j b_j^2}\,\sqrt{\mathbb{E}_j(1 + \|u_j\|/\sqrt{Q})^2} \leq \sqrt{\mathbb{E}_j b_j^2}\,(1+r). \end{aligned}$$

The last step uses $\mathbb{E}_j\|u_j\| \leq \sqrt{\mathbb{E}_j\|u_j\|^2}$. Therefore $E_N \geq dQr^2 + 1/[4(1+r)^2]$. The displayed clones attain this bound: every active signed preactivation is $r > 0$, and every other one is at least r.

The scalar objective has derivative $2dQr - 1/[2(1+r)^3]$ and strictly positive second derivative. Its derivative at zero is negative, so its unique minimum is at $r > 0$ and below its value $1/4$ at zero. For $Q_1 > Q_0$, evaluating the profile for Q_0 at the positive minimizer r_1 for Q_1 gives

$$\Psi_d(Q_1) - \Psi_d(Q_0) \geq d(Q_1 - Q_0)r_1^2 > 0.$$

□

The theorem is an inequality over all neurons and all nonlinear output functions. It does not restrict competitors to cloned networks. For a threshold truth table g, apply it to the complete labeled cube: if $Q(g)$ is the minimum affine unit-margin squared norm, then the strict-margin cell has energy infimum $\Psi_d(Q(g))$. A small common readout amplification makes the cloned margins strict.

Lemma 3 (The nonthreshold gap for the linear residual)**.** *If a Boolean function g is not an affine threshold function, then every network satisfying $g(x)F(x) \geq 1/2$ on the full cube has $E_N \geq 1/4$.*

Proof. Put $z_x = g(x)\bar{x}$. Since these finitely many vectors do not admit a strict separating linear functional, the origin belongs to their convex hull. Indeed otherwise strict separation of the origin from that compact convex hull would give an affine threshold representation of g. Choose probabilities ρ_x with $\sum_x \rho_x z_x = 0$. Then

$$S(u) := \sum_x \rho_x \phi(z_x \cdot u) = \sum_x \rho_x \operatorname{sign}(z_x \cdot u), \qquad |S(u)| \leq 1.$$

Averaging the margins gives $1/2 \leq \mathbb{E}_j[b_j S(u_j)] \leq \mathbb{E}_j|b_j| \leq \sqrt{\mathbb{E}_j b_j^2}$. The readout part of the energy alone is therefore at least $1/4$. □

3 The same exact reduction with a bounded residual

The bounded activation gives the same online separation without an unbounded output from a hidden neuron. We prove a slightly more general statement for

$$\phi_c(t) = \operatorname{sign}(t) + c\tanh(t), \qquad 0 < c \leq 1.$$

The coefficient c is fixed across input dimensions and learner widths.

Theorem 4 (A bounded threshold residual)**.** *For any nonempty linearly separable labeled subset of the Boolean cube, let a_*, Q be its affine unit-margin minimizer and squared norm as above. At every width, the minimum training-margin energy for ϕ_c is*

$$\Xi_{d,c}(Q) = \min_{r \geq 0} \left\{ dQr^2 + \frac{1}{4(1 + c\tanh r)^2} \right\}. \tag{5}$$

The unique minimizing radius is positive and satisfies

$$4dQr(1 + c\tanh r)^3 = c\operatorname{sech}^2 r. \tag{6}$$

Cloning $u_j = ra_*$ and $b_j = 1/[2(1 + c\tanh r)]$ attains the minimum. The profile is strictly increasing in Q and is strictly below $1/4$. Every globally nonthreshold Boolean output still has energy at least $1/4$.

Proof. Every signed augmented input has norm $\sqrt{d+1}$. The unit-margin constraint at any training point implies $Q \geq 1/(d+1)$, so $dQ \geq 1/2$. The scalar objective in (5) has negative derivative at zero and strictly increasing derivative: its negative derivative term is proportional to $c\operatorname{sech}^2 r/(1 + c\tanh r)^3$, which decreases strictly. Thus its minimizer is unique and positive and satisfies (6). In particular $r \leq c/(4dQ) \leq 1/2$.

Write

$$h = 1 + c\tanh r, \quad \alpha = 1 + c(\tanh r - r\operatorname{sech}^2 r), \quad \beta = c\operatorname{sech}^2 r, \quad \lambda = h^{-2}, \quad K = \lambda^2/4.$$

Then $h = \alpha + \beta r$, $\alpha \geq 1$, $0 < \beta \leq 1$, and $K \leq 1/4$. For any score $s \in [-\sqrt{2}, \sqrt{2}]$,

$$-\alpha + \beta s \leq \phi_c(s) \leq \alpha + \beta s. \tag{7}$$

For $s > 0$, the upper inequality is the concave tangent bound for tanh at r. For $s < 0$, $\phi_c(s) \leq -1$, whereas $\alpha + \beta s \geq 1 - \sqrt{2} > -1$. At zero it is immediate. Oddness gives the lower inequality. The fact $\alpha \geq 1$ follows from $\tanh r - r \operatorname{sech}^2 r \geq 0$, whose derivative is $2r \operatorname{sech}^2 r \tanh r \geq 0$.

Use the active KKT probabilities ρ_i and put $S(u) = \sum_i \rho_i \phi_c(z_i \cdot u)$. For any atom and any readout b, completing the square gives

$$d\|u\|^2 + b^2 - \lambda b S(u) \geq d\|u\|^2 - K S(u)^2.$$

If $\|u\| \leq 1/\sqrt{d}$, every score has magnitude at most $\sqrt{2}$. Equation (7) and the KKT mean give

$$|S(u)| \leq \alpha + \beta \|u\| / \sqrt{Q}.$$

Writing $t = \|u\|/\sqrt{Q}$, the atomwise lower bound is therefore at least

$$\min_{t \geq 0} \{dQt^2 - K(\alpha + \beta t)^2\} = dQr^2 - Kh^2 =: \zeta.$$

Indeed $dQ - K\beta^2 \geq 1/4 > 0$, and stationarity says $dQr = K\beta h$, making $t = r$ its unique minimum. It also gives $\zeta = -\alpha/(4h^3) < 0$. If $\|u\| \geq 1/\sqrt{d}$, boundedness gives instead

$$d\|u\|^2 - K S(u)^2 \geq 1 - 4K \geq 0 > \zeta.$$

Thus the same floor ζ holds for every atom. Averaging and using the training margins yields

$$E_N \geq \lambda/2 + \zeta = dQr^2 + \frac{1}{4h^2}.$$

The proposed clones attain equality on the active constraints and satisfy all others by monotonicity of ϕ_c on positive scores. Strict monotonicity in Q follows by evaluating the profile at the positive minimizer for the larger Q, as in Theorem 2. Its value is below $1/4$ because its derivative at zero is negative.

For the final assertion, choose the zero-mean convex-hull probabilities for a nonthreshold function as in Lemma 3. The tangent estimate (7) at $r = 0$ has $\alpha = 1, \beta = c$. Thus $|S(u)| \leq 1$ on $\|u\| \leq 1/\sqrt{d}$. On the remaining atoms $|S(u)| \leq 1 + c \leq 2$. With square-completion coefficient one,

$$d\|u\|^2 + b^2 - bS(u) \geq d\|u\|^2 - S(u)^2/4 \geq -1/4$$

in both regions; in the large-atom region the right side is actually nonnegative. Averaging and using $\mathbb{E}_j b_j S(u_j) \geq 1/2$ proves $E_N \geq 1/4$. □

From now on write $J_d(Q)$ for $\Psi_d(Q)$ with the linear residual, or $\Xi_{d,1}(Q)$ with the bounded residual. Both profiles are strictly increasing and below $1/4$, and both have explicit cloned minimizers. The same arguments also cover every fixed $0 < c \leq 1$ in the bounded case, with $J_d = \Xi_{d,c}$.

4 A fixed input order with exponentially many affine mistakes

We give the complete affine construction used by the network theorem. Keep the target f_k from Theorem 1, and abbreviate the two encoded integers by X and Y. The calibration examples all have selector $z = -1$.

1. Take diagonal inputs $x = y$ equal to the all-positive vector, each of its k one-coordinate flips, and the negatives of these vectors. Every label is positive.

2. For each $i \in \{0, \dots, k-1\}$, take two negative examples with $x_i = -1, y_i = +1$ and $x_j = +1, y_j = -1$ for $j < i$. For $j > i$, set the equal bits $x_j = y_j$ all positive in one example and all negative in the other. Their highest differing bit is i, so $X < Y$.

Remove duplicate examples. The resulting calibration has at most $4k + 2$ inputs. Let q_t denote the input $z = +1$, $Y = 0$, $X = t$; its target label is negative for every $0 \le t \le 2^k - 1$.

Lemma 5. *For $k \le t \le 2^k - 1$, the unique minimum-norm affine unit-margin coefficient vector fitting the calibration and the negative example q_t is the vector of the affine score*

$$a_t \cdot \bar{x} = -t - (t+1)z + \sum_{i=0}^{k-1} 2^i (x_i - y_i). \tag{8}$$

It fits all earlier q_j with $j \le t$, and its absolute score is at least one on the entire cube. Its squared norm is

$$Q_t = t^2 + (t+1)^2 + 2\sum_{i=0}^{k-1} 4^i. \tag{9}$$

For $t < 2^k - 1$, it predicts q_{t+1} positively, and adding the correct negative label raises the convex affine minimum by

$$Q_{t+1} - Q_t = 4(t+1) > 0. \tag{10}$$

Proof. Write an arbitrary affine score as

$$b + cz + \sum_i a_i x_i + \sum_i h_i y_i,$$

and set

$$u = b - c, \quad v = b + c, \quad w_i = (a_i - h_i)/2, \quad s_i = a_i + h_i, \quad U = u - 1.$$

The squared coefficient norm is

$$\frac{u^2 + v^2}{2} + 2\sum_i w_i^2 + \frac{1}{2}\sum_i s_i^2. \tag{11}$$

The paired positive diagonal constraints give

$$U \ge 0, \qquad \left|\sum_j s_j\right| \le U, \qquad \left|\sum_j s_j - 2s_i\right| \le U.$$

Subtracting the expressions inside the last two absolute values shows $|s_i| \le U$, hence $\sum_i |s_i| \le kU$. Average the two negative carry constraints at coordinate i. The upper equal-bit terms cancel and leave

$$u - 2w_i + 2\sum_{j<i} w_j \le -1.$$

Induction yields

$$w_i \geq \sum_{j<i} w_j + \frac{u+1}{2} \geq 2^i(1 + U/2). \tag{12}$$

At q_t, the non-selector part of the input contribution is $2\sum_{\mathrm{bit}_i(t)=1} w_i - \sum_{\mathrm{bit}_i(t)=0} s_i$. Its negative unit-margin constraint therefore implies

$$v \leq -1 - 2\sum_{\mathrm{bit}_i(t)=1} w_i + \sum_{\mathrm{bit}_i(t)=0} s_i \leq -(2t+1) + (k-t)U \leq -(2t+1),$$

since $t \geq k$. Combining this with $u \geq 1$ and (12) in (11) gives

$$\|a\|^2 \geq \frac{1 + (2t+1)^2}{2} + 2\sum_i 4^i = Q_t.$$

Equality requires $u = 1$, $s_i = 0$, $w_i = 2^i$ and $v = -(2t+1)$, which is exactly (8).

Conversely the displayed score equals $1 + 2(X - Y)$ on $z = -1$, and $-(2t+1) + 2(X - Y)$ on $z = +1$. It fits every calibration label, and its score at q_j is $2j - 2t - 1 \leq -1$ for $j \leq t$. Both branch scores are odd integers on the full cube, so the absolute margin is at least one. At q_{t+1} the score is $+1$. Applying the same minimum calculation to the augmented history gives Q_{t+1}, and subtraction gives (10). □

The input permutation starts with the calibration, then q_k, then $q_{k+1}, \ldots, q_{2^k-1}$, and finally every remaining input in lexicographic order. These blocks are disjoint. Its calibration prefix has length at most $4k + 3$, and there are $T_k = 2^k - k - 1 \geq 2^{k-1}$ designated queries. Just before each one, the wrong affine witness has the exact convex margin minimum for all data seen so far.

The target has a linear-size Boolean circuit. With 0/1 bit notation, comparison of the lowest $i + 1$ bits obeys

$$G_{-1} = 1, \qquad G_i = (X_i \wedge \neg Y_i) \ \vee \ ((X_i \leftrightarrow Y_i) \wedge G_{i-1}).$$

Each step uses a constant number of bounded-fan-in gates; conjoin the last result with $z = -1$. Every input bit is essential: for x_i use $Y = 2^i$ and all other X bits zero; for y_i use $X = 0$ and all other Y bits zero; for z take $X = Y$. A fan-in-two circuit with d essential inputs has at least $d - 1$ gates, by counting edges in the connected ancestry of its output. This proves the matching $\Theta(k)$ bound.

5 The posterior comparison, including growing width

The following estimate handles the discontinuity by using a neighborhood whose preactivations stay away from zero.

Lemma 6 (A product-box Gaussian comparison)**.** *Suppose a fixed finite history has a cloned single-neuron witness of energy e_0 with a full-cube nonzero affine preactivation at every input, and whose output has the desired global weak margin and history labels. Suppose every network in a competing event has energy at least $e_1 > e_0$, at every width. Then there are constants $\delta > 0$ and $C \geq 0$, independent of N and σ, such that the competing posterior mass is at most*

$$\exp\left[-\frac{\delta}{2\sigma^2} + CN\right] \tag{13}$$

whenever $(d+2)N\sigma^2 < e_1$. The posterior is conditioned on the history and global margin.

Proof. In physical neuron coordinates

$$\theta_j = (\sqrt{d}\, u_j, b_j) \in \mathbb{R}^q, \qquad q = d + 2,$$

the rows are independent centered Gaussians with covariance $s^2 I_q$, where $s^2 = N\sigma^2$, and $E_N = N^{-1} \sum_j \|\theta_j\|^2$. Amplify the witness's readout slightly, so every margin is strict and its energy remains below e_1. Its preactivations are nonzero on the finite cube. Consequently there is a fixed open coordinate box $B \subset \mathbb{R}^q$ of volume $V > 0$ about this amplified neuron such that every neuron in B has the same strict full-cube sign function and individual signed output greater than $1/2$, and

$$R_B := \sup_{\theta \in B} \|\theta\|^2 < e_1.$$

This uses continuity only within a region of fixed nonzero preactivation signs. If every neuron lies in B, their average satisfies the same strict margins and history. Thus the posterior denominator is at least

$$\left[V (2\pi s^2)^{-q/2} e^{-R_B/(2s^2)} \right]^N.$$

For the numerator, NE_N/s^2 is chi-square with Nq degrees of freedom. The exponential-moment bound, valid for $e_1 > qs^2$, gives

$$\log \mathbb{P}(E_N \geq e_1) \leq -\frac{Ne_1}{2s^2} + \frac{Nq}{2} \left[1 + \log \frac{e_1}{qs^2} \right].$$

Dividing by the denominator cancels all logarithmic s^2 terms. With $\delta = e_1 - R_B > 0$, the logarithm of the ratio is at most

$$-\frac{\delta}{2\sigma^2} + N \left\{ \frac{q}{2} \left[1 + \log \frac{2\pi e_1}{q} \right] - \log V \right\}.$$

Replace the expression in braces by its maximum with zero to obtain (13). □

Before the designated query q_{t+1} in Lemma 5, the cloned witness has energy $e_0 = J_d(Q_t)$, has no zero affine score on the cube, and predicts that query positively. Every network making the correct negative prediction there belongs to a global-margin cell and therefore satisfies the enlarged training set with margins at least $1/2$. Theorems 2 and 4 bound its energy by $e_1 = J_d(Q_{t+1}) > e_0$. Lemma 6 applies directly. There is no appeal to continuity of the activation at zero or to a compactness argument across its jumps.

For each fixed N, (13) tends to zero as $\sigma \downarrow 0$. Under (2), it is

$$\exp[-\delta N^\gamma/(2\tau^2) + CN].$$

For every fixed $\gamma > 1$ and $\tau > 0$ this tends to zero; the tail condition in Lemma 6 also holds eventually. For $\gamma \geq 1$ and sufficiently small τ, both the tail condition and any desired bound on the competing probability hold at *every* $N \geq 1$: require

$$\tau^2 < \frac{e_1}{d+2}, \qquad \frac{\delta}{2\tau^2} - C \geq \log(T_k/\eta).$$

Taking common constants over the finite list of designated histories and using the union bound proves all online assertions of Theorem 1. Independence between posterior samples at different rounds is unnecessary.

6 A width-one teacher and polynomial IID learning

The final affine witness a_{2^k-1} represents the target on the entire cube: on $z = -1$ its score is $1+2(X-Y)$, and on $z = +1$ it is always negative. Multiply all its affine coefficients by a sufficiently small positive number so that every one has magnitude at most one, and use readout weight one. All preactivations remain nonzero and retain their signs. Both activations have $|\phi(t)| > 1$ for $t \neq 0$, so the resulting width-one teacher has signed output greater than one. The linear Boolean circuit and its matching lower bound were given with the selector construction.

Theorem 7 (IID guarantee for the same prior)**.** *Fix any affine threshold target f on $\{-1,1\}^d$, any input distribution $\mathcal{D}$, and $\varepsilon, \beta \in (0,1)$. If*

$$m \geq \frac{4}{\varepsilon}\left[(d+1)^2 \log 2 + \log(1/\beta)\right],$$

then with probability at least $1-\beta$ over m IID examples labeled by f, the posterior correct-label probability exceeds $2/3$ outside a set of $\mathcal{D}$-mass at most ε in each of these regimes:

1. *at every fixed width, for all sufficiently small $\sigma > 0$;*
2. *under (2), at every sufficiently large width for any fixed $\gamma > 1$ and $\tau > 0$;*
3. *at every width and every $\gamma \geq 1$ in (2), when $\tau > 0$ is sufficiently small.*

The variance and width thresholds can be chosen independently of the training sample and $\mathcal{D}$.

Proof. The full target has a cloned witness of energy $e_f = J_d(Q(f)) < 1/4$, with nonzero full-cube affine scores. Lemma 3 and Theorem 4 put every nonthreshold output cell in the event $E_N \geq 1/4$. The full-target witness is consistent with every sample labeled by f. Lemma 6, with $e_1 = 1/4$, therefore bounds the total posterior mass of all nonthreshold functions uniformly over every such history. This bound tends to zero in the first two regimes and can be made at most $\varepsilon/12$ uniformly in the third. No count of nonthreshold functions is needed.

Let $\mathcal{H}_d$ be the threshold truth tables on the cube. The 2^d hyperplanes in the $(d+1)$-dimensional coefficient space give

$$|\mathcal{H}_d| \leq \sum_{j=0}^{d+1} \binom{2^d}{j} \leq (2^d+1)^{d+1} \leq 2^{(d+1)^2}.$$

A consistency union bound shows that, with probability at least $1-\beta$, every sample-consistent threshold function has population error at most $\varepsilon/4$. On this sample event, posterior mean population error is at most $\varepsilon/4 + \varepsilon/12 = \varepsilon/3$ once the nonthreshold mass is small enough. An input with correct-label posterior probability at most $2/3$ has wrong-label probability at least $1/3$. Markov's inequality bounds the mass of those inputs by ε. □

7 What the activation contributes

The threshold component supplies a signed output of magnitude one even when its incoming weights are arbitrarily small. Either residual then makes a small positive incoming scale strictly preferable to zero: its energy optimum is below $1/4$ and depends strictly on the affine margin norm. This creates an energy order at every width while keeping a width-one bounded teacher for every affine threshold target.

The elementary linear-residual calculation also works for every fixed $\operatorname{sign}(t) + ct$ with $c > 0$, replacing $1 + r$ by $1 + cr$ in its profile. For the bounded family, Theorem 4 already permits every fixed $0 < c \leq 1$. Its uniform distance from pure sign is at most c. The residual may therefore be arbitrarily small, but its coefficient is fixed across widths and dimensions. At $c = 0$ the strict energy order disappears, so that limiting case needs a different argument.

This example uses an unchanged Gaussian prior and actual posterior conditioning, not a favorable choice of a minimum-energy predictor. Its activation is discontinuous. Replacing the jump by a fixed continuous transition is a substantive mathematical change, and the present proof does not justify that replacement. The question for ordinary tanh, ReLU, or hardtanh under this same prior remains separate.